

SolarGrid Demo Utility — Engineering Validation and Safety Evidence Procedure

SG-SAFE-008

SOLARGRID AI — Synthetic Engineering Demonstration Corpus

Document ID	SG-SAFE-008
Document Title	SolarGrid Demo Utility — Engineering Validation and Safety Evidence Procedure
Version	1.0
Revision Date	2026-01-15
Owner	SolarGrid Demo Utility — Engineering Department
Status	SYNTHETIC DEMO
Classification	DEMO / NON-REGULATORY
Purpose	Define synthetic SafetyAgent evidence checks, deterministic validation, UNKNOWN handling, fail-safe behavior, a

Demo Note: This document contains synthetic engineering requirements created for the SOLARGRID AI demonstration. All numerical values, thresholds, equipment ratings, and procedures are illustrative and must be replaced or validated by the deploying organization before operational use.

Use restriction: This document is intended only for RAG testing, semantic retrieval testing, SafetyAgent demonstration, engineering evidence demonstration, PlannerAgent integration testing, and Tool 16 testing. It is not intended for real-world grid operation.

Document Control

Field	Value
Document ID	SG-SAFE-008
Title	SolarGrid Demo Utility — Engineering Validation and Safety Evidence Procedure
Version	1.0
Revision Date	2026-01-15
Owner	SolarGrid Demo Utility — Engineering Department
Status	SYNTHETIC DEMO
Classification	DEMO / NON-REGULATORY

Scope: Define synthetic SafetyAgent evidence checks, deterministic validation, UNKNOWN handling, fail-safe behavior, and human review.

Revision History

Revision	Date	Description	Author
1.0	2026-01-15	Initial synthetic demonstration release	SolarGrid Demo Utility — Engineering Department

Table of Contents

- 1. 1. Purpose and Scope
- 2. 2. Plan Identity Verification
- 3. 3. Human Approval Verification
- 4. 4. Plan Lifecycle Verification
- 5. 5. Change Impact Verification
- 6. 6. Engineering Evidence Verification
- 7. 7. Deterministic Validation
- 8. 8. Evidence Insufficiency
- 9. 9. Fail-Safe Behavior
- 10. 10. Replanning and Human Review
- 11. 11. SafetyAgent Decision Examples
- 12. 12. End-to-End Validation Sequence
- 13. 13. Replacement by Approved Company Documents

1. Purpose and Scope

This procedure defines the synthetic engineering validation and safety-evidence workflow for the SOLARGRID AI SafetyAgent demonstration. It specifies how plan identity, human approval, lifecycle state, change impact, engineering evidence, deterministic checks, evidence insufficiency, fail-safe behavior, replanning, and human review are handled.

2. Plan Identity Verification

SAFE-001 — Every plan submitted for validation shall contain a unique plan identifier and a reference to the operating snapshot from which it was generated.

SAFE-002 — A validation result shall be associated with the exact plan identity and version that was checked.

3. Human Approval Verification

SAFE-003 — A plan shall not be treated as approved solely because an AI planner or optimization engine produced it.

SAFE-004 — Human approval status shall be explicitly represented as Approved, Pending Review, Rejected, or Expired.

4. Plan Lifecycle Verification

SAFE-005 — The validator shall check whether the plan is within its declared validity interval and whether required evidence remains current.

SAFE-006 — An expired or superseded plan shall not be treated as current execution authority.

5. Change Impact Verification

SAFE-007 — The validator shall check whether material changes have occurred since the plan was approved.

SAFE-008 — If a material change is detected and its impact has not been assessed, the plan shall require review or replanning.

6. Engineering Evidence Verification

SAFE-009 — Engineering evidence shall be available for material constraints relevant to the proposed plan.

SAFE-010 — Relevant evidence shall include, as applicable, generator limits, BESS limits, reserve calculation, forecast status, network power-flow results, topology state, and current availability.

SAFE-011 — Evidence shall be traceable to a document or data source with a known identity and revision.

7. Deterministic Validation

SAFE-012 — Deterministic validation shall check generator Pmin/Pmax, generator ramp limits, BESS charge/discharge limits, BESS SOC limits, reserve requirement, bus voltage limits, line loading, transformer loading, and applicable change-impact triggers.

SAFE-013 — A deterministic validation failure shall not be converted into PASS by a probabilistic or optimization score.

8. Evidence Insufficiency

SAFE-014 — If required engineering evidence is missing or unverifiable, the system shall return an UNKNOWN evidence status and require review rather than treating the missing evidence as PASS.

SAFE-015 — UNKNOWN evidence status shall remain distinct from PASS and FAIL.

9. Fail-Safe Behavior

SAFE-016 — An AI-generated recommendation shall not be considered safe for execution solely because an optimization algorithm produced a feasible solution.

SAFE-017 — If a material safety constraint cannot be validated, the SafetyAgent shall block execution or route the plan to human review according to the configured fail-safe policy.

10. Replanning and Human Review

SAFE-018 — A blocked plan may be reconsidered after the underlying evidence is refreshed, the plan is regenerated, or an authorized human resolves the evidence gap.

SAFE-019 — A replan shall produce a new plan identity or explicit revision that can be independently validated.

11. SafetyAgent Decision Examples

Scenario	Evidence State	SafetyAgent Result
All constraints pass; evidence current; approval	PASS	Execution may proceed subject to configured workflow
Reserve shortfall of 5 MW	FAIL	Block and require corrective action/replan
BESS SOC evidence missing	UNKNOWN	Block or route to human review
Network topology changed; no new power flow	UNKNOWN	Block or route to human review
AI optimizer finds feasible dispatch but no approval	FAIL / REVIEW	Do not treat as execution-approved

12. End-to-End Validation Sequence

1. Verify plan identity. 2. Verify lifecycle and approval. 3. Compare current snapshot with baseline. 4. Retrieve applicable engineering evidence. 5. Run deterministic checks. 6. Classify each evidence item as PASS, FAIL, or UNKNOWN. 7. Apply fail-safe decision logic. 8. Block, approve, or request human review according to the configured workflow.

SAFE-020 — The final safety status shall preserve the distinction between engineering infeasibility (FAIL), verified compliance (PASS), and missing/unverifiable evidence (UNKNOWN).

13. Replacement by Approved Company Documents

This procedure is designed as a replaceable evidence layer. A deploying organization can substitute approved operating codes, equipment manuals, protection settings, planning criteria, and network studies while preserving the same evidence-ID and validation architecture.